

Greek fortition in Logical Phonology

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Logical Phonology (henceforth, LP; e.g., Bale et al. 2014, Bale and Reiss 2018, Dabbous et al. 2021, Gorman and Reiss 2024, Reiss 2021) is a formally rigorous, *substance-free* theory of phonology. LP posits that features are binary-valued (rather than privative) and segments are thus sets of valued features linked to a X-slot. A segment must be *consistent*: if a segment ς is specified +F, it cannot also be specified –F or vice versa. However, a segment can be *underspecified*: ς need not contain any specification for F at all. Following Poser (1982) et seq., the distinction between feature-filling and feature-changing rules is derived by decomposing feature-changing processes (e.g., voice assimilation) into separate, independently motivated feature insertion (*unification*) and deletion (*subtraction*) operations.

(1) Unification ($\sqcup$): If A and B are feature specifications, and their union $A \cup B$ is consistent, then $A \sqcup B = A \cup B$; otherwise, $A \sqcup B = A$.

(2) Subtraction ($\setminus$): If A and B are feature specifications, then $A \setminus B = \{cF \mid cF \in A \wedge cF \notin B\}$.

Unification provides a novel solution to the decades-old problem (e.g., Archangeli 1988, Bonet 1995, Lees 1961:12–14, Lightner 1971:236) of targeting relatively underspecified structures without also affecting more richly specified ones. It is also a principled way to implement Inkelas and colleagues’ (e.g., Inkelas and Cho 1993, Inkelas 1995, Inkelas and Orgun 1995, Inkelas et al. 1997) suggestion that segments underspecified for some feature F are *mutable* with respect to processes inserting $\pm F$, allowing many patterns which were previously analyzed using *morphophonological* rules, morpheme-specific constraints, cophonologies, or lexical diacritic features to be generated by the narrow phonology.

Markopoulos and Revithiadou (2023), henceforth M&R, describe a process of manner dissimilation in Modern Greek by which certain labial fricatives, those spelled φ , β as in (3ab),

strengthen—and devoice in the case of β —to [p] before the sibilant /s/; they are realized as [f, v] elsewhere. However, some labial fricatives, spelled υ as in (3c), never undergo fortition.

		IPFV.1SG /-o/	PFV.1SG /-s-o/	
(3)	a.	γράφ-	[ˈɣrafo]	[ˈɣrapso] ‘write’
	b.	ράβ-	[ˈravo]	[ˈrapso] ‘sew’
	c.	βραβεύ-	[vraˈvevo]	[vraˈvefso] ‘award’

M&R cite a number of acoustic studies confirming these transcriptions and conduct a formal judgment task ($n = 35$); they then analyze this pattern using Gradient Harmonic Grammar (Smolensky and Goldrick 2016). I now show it can be generated using the more modest architectural assumptions of LP. I propose that *inalterable* $\langle \upsilon \rangle$ /v/ is underlyingly +CONTINUANT /v/, whereas *mutable* $\langle \phi, \beta \rangle$ /F, V/ are underspecified for this feature.

(4) Feature specification (partial):

	$\langle \phi \rangle$	$\langle \beta \rangle$	$\langle \upsilon \rangle$
	/F/	/V/	/v/
CONTINUANT			+
VOICE	–	+	+

(5), a unification rule strengthens /F, V/ when followed by a sibilant, but applies vacuously to inalterable /v/. A later context-free redundancy rule specifies /F, V/ not targeted by (5) as fricatives. For completeness, I also provide voice assimilation rules and sample derivations.

(5) Fortition: $\left[+\text{LABIAL} \right] \sqcup \left\{ -\text{CONTINUANT} \right\} / - \left[+\text{STRIDENT} \right]$

(6) Default frication: $\left[+\text{LABIAL} \right] \sqcup \left\{ +\text{CONTINUANT} \right\}$

(7) Voice assimilation (part 1: deletion): $\left[-\text{SONORANT} \right] \setminus \left\{ \alpha\text{VOICE} \right\} / - \left[\alpha\text{VOICE} \right]$

(8) Voice assimilation (part 2: insertion): $\left[-\text{SONORANT} \right] \sqcup \left\{ \alpha\text{VOICE} \right\} / - \left[\alpha\text{VOICE} \right]$

(9) Critical orderings: (5) $\ll$ (6); (7) $\ll$ (8)

(10) Sample derivations (stress omitted):

UR	/yraF-o/	/yraF-s-o/	/raV-o/	/raV-s-o/	/vravev-o/	/vravev-so/
Rule (5):		yrapso		rabso		
Rule (6):	yrafo		ravo			
Rules (7–8):				rapso		vravefso
SR	[yrafo]	[yrapso]	[ravo]	[rapso]	[vravevo]	[vravefso]

LP provides all necessary representations and operations needed to implement Greek’s “exceptional” fortition pattern, analyzed by M&R using the much richer architectural primitives of Gradient Harmonic Grammar.

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